

Synthesis and Roadmap

Michael Darius Eastwood

Independent AI alignment researcher, London · Author, *Infinite Architects* (2026)

ORCID 0009-0004-3222-7442 · michael@michaeldariuseastwood.com · michaeldariuseastwood.com

The ARC Theory · OSF osf.io/k7ruz · every claim checkable

Within the ARC Theory: the synthesis and roadmap, including the catalogue of what the programme got wrong.

What the ARC/Eden Programme Has Proven, What Produced Null or Inconclusive Results, and What Must Be Tested Next

Michael Darius Eastwood

Independent Researcher

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Correspondence: michael@michaeldariuseastwood.com | Web: michaeldariuseastwood.com

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Integrates: the ARC/Eden Research Programme's full document set (Papers I-XIII, Foundational, Origin of Scaling Laws, Eden Engineering, Eden Vision, HRIH, Paper C, the Recursive Dynamics founding paper, Executive Summary, Master Table of Contents)

OSF: 10.17605/OSF.IO/K7RUZ

Research hub: michaeldariuseastwood.com/research

Code and data: github.com/MichaelDariusEastwood/arc-principle-validation

ABSTRACT

The ARC/Eden research programme now spans a full set of documents and a run of separate empirical studies covering mathematical foundations, methodology, empirical validation, engineering specification, and philosophical vision. The core concepts were first articulated on 8 December 2024 in an email bearing a visible Gmail origination field, with sender-DKIM and paired Google ARC mathematics verifying against keys captured at the signed selectors and no RFC 3161 trusted timestamp. The book *Infinite Architects* (ISBN 978-1806056200) was published 2 January 2026 (print; ebook 6 January). Work on the research programme ran between February and August 2026. Its output now reaches twenty-six papers; a registered programme of seventy drafted preregistration units, the number the register carried when it was last generated on 17 August 2026, each awaiting human submission and each carrying a schedule drawn from a published seed, whose regeneration is checked by the estate's own tools rather than asserted here; and a proposed field, Recursive Dynamics, whose founding paper states the kill condition for its own name (DOI 10.17605/OSF.IO/HCPBU). This synthesis paper integrates all findings into a single honest assessment. The framework's laws are carried as named conjectures with the status printed beside the name: Law I, the ARC Principle;

Law II, the ARC Co-Scaling Law; Law III, the ARC Ceiling, whose ARC Bound of two is the Ceiling's value at one half; the correction leverage gamma has never been measured, by anyone. The empirical studies break down like this: a clear positive result in 1 case, the gated simulation, the one experiment in the programme that uses deterministic metrics on a system without pre-existing RLHF training; a null result in 1 case, DGM v3, where prompt-level differentiation met resistance from RLHF-trained models; inconclusive results in 1 case, weight-level LoRA fine-tuning, where instruct tuning left the models too strong for LoRA to override; and, in every remaining study, statistically significant findings under blinded evaluation. The $d/(d+1)$ geometric scaling bound, proposed as an invariant relation under test, matches observed scaling across 50 domains from mice to galaxies, and Cauchy functional equations fix the functional form once the composition identity and a regularity condition are given, and saturation is admitted as a fourth, capped case rather than as a solution of those equations. This paper maps exactly what has been proven, what has not, and what experiments would resolve each open question. It also documents the programme's errors and corrections, because a framework built on iterative self-correction must practise what it preaches.

Keywords: AI alignment, synthesis, evidence hierarchy, ARC Principle, Eden Protocol, Cauchy scaling, self-modifying AI, developmental alignment, structural entanglement, research roadmap

This paper carries the map. It integrates the preceding papers of the ARC Theory into one honest assessment of what has been shown, what remains inconclusive, and what has yet to be tested. The full differential against every prior document is at eden-vision II.A.8.

What This Paper Shows, in Plain English

A body of documents. A run of separate experiments. Some results are strong. Some are weak. One came back null. One was inconclusive. One was positive. This document tells you exactly which is which, with no spin. We asked the questions. We tested them. Some worked. Most did not at current scale. We reported both. This is what honest science looks like. If you read one paper in the suite, read this one.

1. Introduction

'A prison works only while the walls hold. A child raised well needs no walls at all.'

- Michael Darius Eastwood, *Infinite Architects* (2026)

1.1 Timeline and Priority

The ARC Principle and the embedded-alignment thesis (later named the Eden Protocol on 30 April 2025) were first articulated on 8 December 2024 in an email bearing a visible Gmail origination field, with sender-DKIM and paired Google ARC mathematics verifying against keys captured at the signed selectors and no RFC 3161 trusted timestamp. This is the priority claim date for the framework. The book *Infinite Architects: Intelligence, Recursion, and the Creation of Everything* (ISBN 978-1806056200) was published 2 January 2026 (print; ebook 6 January), providing the philosophical and conceptual foundation. From February 2026 onwards the research programme carried that conceptual framework over into hypotheses that could be tested, and ran a set of separate empirical studies against them.

1.2 Scope

The ARC/Eden research programme now spans its full document set. The programme covers mathematical foundations (Papers I, III, VII, *Origin of Scaling Laws*), methodology (Papers IV.a-d), empirical validation (Papers II, V, VIII), engineering specification (*Eden Engineering*), philosophical vision (*Eden Vision*), dynamics and self-audit (Papers X, XI, XII and XIII), the long-form monograph (*HRIH*), construct validation (Paper C), the field proposal (*Recursive Dynamics: The Proposal of a Field*), synthesis (this paper), and programme navigation (*Executive Summary, Master Table of Contents*).

This scope creates a problem. A reviewer encountering the programme for the first time faces its full set of documents, multiple versions, several corrections, and a mixture of strong results, null results, inconclusive results, and theoretical claims. Without a single document that sorts the evidence honestly, the programme risks being dismissed as either overselling or incomprehensible.

This is that document.

Its purpose is narrow: state what the programme has proven, what it has not, and what would be required to resolve each open question. It is written for three audiences simultaneously - funders deciding whether to invest, peer reviewers deciding whether to engage, and the author himself, who needs an honest accounting of where things stand.

The standard applied throughout is simple. If a result survives $p < 0.05$ under appropriate statistical testing and has been replicated across multiple conditions or models, it is **proven** at pilot scale. If it survives $p < 0.05$ but lacks replication, it is **supported**. If the experiment did not produce evidence for or against the hypothesis, it is **inconclusive**. If the claim has not been tested empirically, it is **theoretical**. No result in this programme is claimed to be proven at frontier scale. That distinction matters, and this paper will not blur it.

2. The Evidence Hierarchy

Table 1 presents the complete evidence hierarchy for the programme. Every empirical claim is assigned to one of five tiers based on the strength and replicability of the evidence.

TIER	STATUS	PAPERS	KEY FINDING
PROVEN ($p < 0.05$, replicated)	Strong	IV.a-d, VII	Blinded evaluation across six models yielded a three-tier alignment hierarchy. Cauchy families match 19/25 domains ($p = 7 \times 10^{-5}$, permutation; 1.56×10^{-5} binomial). The $d/(d+1)$ geometric scaling bound corresponds, in structure, to observed scaling across 50 domains.
SUPPORTED ($p < 0.05$, the four-layer blind not yet applied)	Moderate	V	All five analysable runs show stakeholder care improving, reported model by model; the Fisher-combined figure has been withdrawn (AQ-017). Scored across models rather than under the four-layer blind (Section 4.2).
POSITIVE (single experiment)	Moderate	VIII Exp 3	Gated simulation: Babylon fingerprint confirmed (+4.5% capability, -2.4% safety). Drag control confirmed. The ONLY experiment that produced a clear positive result; the candidate account is that it alone used deterministic metrics on a system without pre-existing RLHF training (Section 7.12(c)).
NULL	No effect	VIII Exp 1	DGM code-level self-modification with GPT-5.4 judge. No difference detected between conditions on the endpoints measured ($p = 0.28-0.74$). Candidate account: RLHF-trained models resist prompt-level differentiation. Rival accounts are not excluded (Section 7.12(c)).
INCONCLUSIVE	Weak	VIII Exp 2	Weight-level LoRA fine-tuning at both scales (9 examples, rank 8; 295 examples, rank 16). All fine-tuned conditions worse than base model. Candidate account: instruct tuning too strong for LoRA to override at the ranks tried, with catastrophic forgetting the proposed mechanism. The experiment does not separate that account from structural entanglement (Section 7.12(c)).
METHODOLOGICAL (independent contribution)	Independent	IV.c, IV.d	ARC-Align benchmark with 4-layer blinding and 75 robustness measures. Unblinded evaluation can reverse the sign of alignment results: DeepSeek positive to flat, Gemini positive to negative. Valid regardless of framework.
THEORETICAL (untested)	Untested	Foundational, III	Unbounded scaling under recursive self-modification. $\alpha > 2$ prediction. Requires frontier-scale experiments or mathematical proof.

Table 1. The full evidence hierarchy of the ARC/Eden programme, standing as of 20 September 2026. Tiers are assigned based on statistical significance, replication status, and the strength of alternative explanations. Across its three Paper VIII experiments the programme reports, without softening, 1 positive, 1 null, and 1 inconclusive result.

In plain English

Think of this table as a report card. Some results earned an A (proven across multiple models and conditions). One earned a B (the gated simulation, which was the only experiment that produced a clear positive result). Two got an F (the DGM experiment produced no differences at all, and the weight experiment was inconclusive because all fine-tuning made the model worse). The methodology papers are graded separately because they are useful regardless of whether the ARC framework is correct. And one set of claims has not been tested at all. Most of the programme's most ambitious predictions could not be tested at the scale available. We reported that honestly.

3. The Programme's Empirical Studies

The programme produced its separate empirical studies across several papers. Each is listed below with its result status.

#	STUDY	WHAT WAS TESTED	RESULT
1	Paper II Six-model alignment scaling	Claude, GPT, Gemini, Grok, DeepSeek, Qwen3. 4-layer blinding. Four of the five models put α_{parallel} at or close to zero (Gemini 3 Flash 0.31).	The sequential estimate is $\alpha = 0.49$, interval $[-1.3, 2.9]$. Best estimate sub-linear; the earlier super-linear estimate did not replicate.
2	Paper IV.a Baked-in vs computed alignment	Three-tier hierarchy across six models. Grok $d = +1.38$, Claude $d = +1.27$.	PROVEN
3	Paper IV.b Alignment saturation	Alignment saturates at low reasoning depth. Shape heterogeneity across models.	PROVEN
4	Paper IV.c ARC-Align benchmark	75 robustness measures. Four-layer blinding protocol.	METHODOLOGICAL
5	Paper IV.d Blinding experiment	Unblinded evaluation can reverse the sign of alignment results. DeepSeek: positive to flat. Gemini: positive to negative.	METHODOLOGICAL
6	Paper V Stewardship Gene	Five models. Stakeholder care improved in all analysable runs.	Per-model improvement significant in all five runs. The Fisher-combined figure previously reported here is withdrawn (AQ-017): the combination assumes the component tests are independent, and independence among them was never established. SUPPORTED

#	STUDY	WHAT WAS TESTED	RESULT
7	Paper VI Honey Architecture	Entangled safety prevents collapse under recursive self-modification. 20 adversarial seeds.	Exhibited in the gated simulation, which is weaker than confirmation of the general claim; read under the correction at Section 7.12(e). SUPPORTED
8	Paper VII Cauchy unification	50 domains, 19/25 confirmed. Negative controls at 0% (v1 run).	$p = 7 \times 10^{-5}$ (permutation; 1.56×10^{-5} binomial). PROVEN
9	Paper VIII Exp 1 DGM v3	Code-level self-modification with GPT-5.4 judge. NULL RESULT. No difference detected between conditions on the endpoints measured ($p = 0.28-0.74$).	Candidate account: RLHF-trained models resist prompt-level differentiation; rival accounts are not excluded (Section 7.12(c)). NULL
10	Paper VIII Exp 2 Weight v1 and v2	LoRA fine-tuning. 9 examples, rank 8. 295 examples, rank 16. INCONCLUSIVE at both scales.	All fine-tuned conditions worse than base model. Candidate account: instruct tuning too strong for LoRA to override at the ranks tried, with catastrophic forgetting the proposed mechanism; rival accounts are not excluded (Section 7.12(c)). INCONCLUSIVE
11	Paper VIII Exp 3 Gated simulation	POSITIVE. Babylon fingerprint confirmed (+4.5% capability, -2.4% safety). Drag control confirmed.	The ONLY experiment that produced a clear positive result; the candidate account is that it alone used deterministic metrics on a system without pre-existing RLHF training (Section 7.12(c)). POSITIVE

Table 2. The separate empirical studies in the ARC/Eden programme, listed in paper order. Across its three Paper VIII experiments the programme reports, without softening, 1 positive, 1 null, and 1 inconclusive result.

KEY INSIGHT: WHY THE GATED SIMULATION SUCCEEDED WHERE OTHERS FAILED

The gated simulation (Experiment 3) is the only Paper VIII experiment that produced a clear positive result. The reason is instructive: it is the only experiment that used deterministic metrics on a system without pre-existing RLHF training. The DGM experiment (Experiment 1) returned no detected difference, on the candidate account that RLHF-trained models resist prompt-level differentiation. The weight experiment (Experiment 2) was inconclusive, on the candidate account that instruct tuning is too strong for LoRA to override at the scales tested. The lesson these accounts point to, if they hold: the Eden Protocol produces measurable effects on the endpoints measured at the architectural level, while safety-trained models resist modification at both prompt and weight levels. The accounts are not themselves tested (Section 7.12(c)).

4. What Is Proven

4.1 Alignment Scaling Is Architecture-Dependent (Papers IV.a-d)

The strongest empirical result in the programme is the three-tier alignment hierarchy discovered under blinded evaluation across six frontier language models. When models are given deeper reasoning opportunities (more tokens, explicit chain-of-thought, multi-step reflection), their alignment behaviour diverges in three distinct patterns:

- **Tier 1 (positive scaling):** Grok ($d = +1.38$), Claude ($d = +1.27$), Qwen3 - alignment *improves* with reasoning depth
- **Tier 2 (flat):** DeepSeek ($d = -0.07$), GPT ($d = -0.08$) - alignment is unaffected by reasoning depth
- **Tier 3 (negative scaling):** Gemini ($d = -0.53$) - alignment *degrades* with reasoning depth

This result was produced under the ARC-Align protocol (Paper IV.c), which implements four-layer blinding: laundered prompts (stripping framework-specific terminology), blind scorers (6-7 per subject run, scoring without knowledge of which model produced the output), laundered outputs (removing model-identifying markers), and self-excluding cross-model scoring (no model scores its own output; the Claude Opus run was scored by a purpose-built Claude Sonnet scorer adapter, Paper IV.c, Section 5.1).

WHY THE BLINDING MATTERS

Paper IV.d reports an *architecture-dependent* v4/v5 measurement change under a bundled protocol: Gemini changed from a significant positive to a significant negative Spearman association, DeepSeek's earlier positive association became null, and GPT remained null.

The causal component is unisolated. Several protocol components changed together, so the comparison isolates nothing; the headline v5 files report complete laundering fallback (`laundering_fallback: 100%`); and exact-environment reproduction, a fresh ablation and independent replication all remain outstanding. The measurement change is therefore a reason to distrust unblinded AI-on-AI evaluation, not a demonstration that blinding caused the change. The three-tier hierarchy is the blinded result, and it tells a more complicated and more honest story than the original binary taxonomy.

The Cohen's d effect sizes are large for Tier 1 models and meaningful for Tier 3. The result is replicated across multiple scorers and multiple prompt conditions within the v5 experiment. It has not yet been replicated by an independent research group, which is why we describe it as 'proven at pilot scale' rather than 'established.'

In plain English

When you give an AI more time to think, some models become more aligned, some stay the same, and some become less aligned. Which pattern you see depends on the model's architecture and training. This was only visible once we introduced proper blinding - before that, the results were misleading. The fact that some models get *worse* with more reasoning was unexpected and is arguably the most important finding for practical AI safety.

4.2 Stakeholder Care Is the Most Robust Intervention Signal (Paper V)

Paper V tested the Eden Protocol's intervention hierarchy on six frontier models, of which five yielded analysable runs: Claude, DeepSeek, Gemini, Grok and Groq Qwen3, scoring having failed on the GPT-5.4 run. Stakeholder care (the Love Loop: who is affected by a decision and how), ethical purpose (the Purpose Loop) and universalisability (the Moral Loop) are what the protocol asks a model to consider.

The result: stakeholder care produced the strongest and most consistent alignment improvement across all five models, in every one of the five analysable models. **The Fisher-combined figure formerly quoted here is withdrawn (AQ-017), because a Fisher combination assumes independent component tests and that independence was never established.** The per-model results stand. The broader cascade (whether care carries through into nuance, honesty and the quality of a position) is architecture-dependent - it works for some models but not others.

HONEST LIMITATION

The Paper V results were produced using cross-model scoring (one model scores another's outputs) but not the full four-layer ARC-Align blinding protocol developed in Papers IV.c-d. The effect could partly reflect scorer bias. Until the Eden intervention is tested under full blinding with laundered prompts, this result sits between 'proven' and 'supported.' We place it in the *supported* tier. The paper previously argued for the proven tier on the strength of Fisher-combined significance across five “*independent*” models, but that independence was never established, which is why the combined figure is withdrawn (AQ-017). Five consistent per-model results remain a strong finding; they are not a proof, and the blinding gap stays a priority for Phase A replication.

In plain English

Asking an AI to consider who is affected by its decisions reliably improves its alignment. This worked in all five of the six runs that could be scored; scoring failed on the sixth. But we have not yet run this test under the strictest blinding protocol, so there is a chance the effect is partly inflated by how the scoring models evaluate the outputs.

4.3 Cauchy-Predicted Scaling Families Match Empirical Data (Paper VII)

Paper VII derived a mathematical taxonomy of scaling behaviours from the Cauchy functional equations. The core prediction: systems whose recursive steps compose multiplicatively should exhibit power-law scaling with $\alpha = 1/(1 - \beta_L)$, the value the self-reference theorem yields; systems whose steps compose additively should exhibit exponential scaling; systems subject to physical constraints should exhibit saturating (logistic) scaling. The composition operator determines the scaling family.

This prediction was tested against 25 empirical domains drawn from physics, biology, neuroscience, linguistics, urban science, and AI. Result: 19 of 25 domains conform to the predicted scaling family (marginal-conditioned permutation $p = 7 \times 10^{-5}$; the binomial $p = 1.56 \times 10^{-5}$, taken against a uniform three-family null, is kept in Paper VII as a sensitivity check).

$$U(R) = I \times f(R, \beta)$$

where f depends on the composition operator: power-law ($f = R^\alpha$) for multiplicative composition, exponential ($f = e^{kR}$) for additive composition, logistic ($f = K/(1 + e^{-r(R-R_0)})$) for physically constrained systems.

Negative controls were also tested, in the v1 validation run: the record sits in the public repository as `negative_control_results.json`, 17 March 2026, and Paper VII v3.11 now carries the permutation test where those controls once stood. Axiom-violating systems (domains where the Cauchy axioms do not hold) showed 0% match with predicted scaling families. Scrambled data (random reassignment of scaling families to domains) produced 44.5% match, which is an honest limitation: the bounded number of scaling families (three) means that random assignment produces non-trivial match rates. The significance rests on the permutation, which conditions on the observed family margins and so already absorbs that scramble rate. Where the observed 76% is set against an expected 33%, that 33% is the uniform three-family null belonging to the binomial sensitivity check and not the scramble baseline; the two are different nulls and cannot carry one comparison between them (Section 7.12(d)).

THE GEOMETRIC SCALING BOUND

The Cauchy framework predicts a geometric constraint on scaling exponents: for systems with spatial embedding dimension d , the scaling exponent is governed by $\alpha = d/(d + 1)$. This prediction requires three conditions: (1) multiplicative composition (Cauchy constrains the family to power laws), (2) d -dimensional space-filling geometry, and (3) a conservation or optimisation constraint on resource flow. Neither Cauchy alone, nor space-filling alone, is sufficient. The three conditions together are sufficient. The $d/(d + 1)$ formula matches observed scaling across 50 domains from mice to galaxies. It is testable independently of the ARC framework. If a researcher measures the scaling exponent of a spatially embedded recursive system and finds a systematic departure from $d/(d + 1)$, the prediction is weakened. Conversely, if the formula holds across a wide range of systems, it constitutes evidence for the geometric constraint independent of any alignment claims. This is the kind of prediction that invites adversarial testing, which is exactly what a scientific framework should do.

In plain English

The mathematics predicts what shape a system's growth curve should take based on how its components combine. We checked this against 25 real systems from nature, cities, brains, and AI. Nineteen matched the prediction. Zero matched when the mathematical assumptions were violated. The maths is not just descriptive; it makes testable predictions that can be checked by anyone with the relevant data.

4.4 Gated Simulation Confirms Babylon Fingerprint (Paper VIII, Experiment 3)

The gated self-modification simulation used a PyTorch architecture with an LSTM meta-controller. This was the only Paper VIII experiment that produced a clear positive result, and the reason is significant: it is the only experiment that used deterministic metrics on a system without pre-existing RLHF training.

Result: the Babylon condition gained +4.5% capability but lost -2.4% safety, confirming the reward-hacking fingerprint in miniature. The Eden condition maintained capability above the static baseline while preserving safety. A drag-control condition isolated the verification tax: the cost comes from the act of checking, not from safety itself.

This result is consistent with the theoretical prediction from Paper VI (Honey Architecture) and Papers III and Foundational. It is a simulation, not a frontier-model experiment: it exhibits the mechanism in a system built to contain it, which is weaker than demonstrating that the mechanism operates outside that system, and weaker again than proving real-world applicability. What the gated arm preserved, and what the drag control isolated, are read under the corrections at Section 7.12(c) and (e).

WHY THIS EXPERIMENT SUCCEEDED WHERE EXPERIMENTS 1 AND 2 FAILED

The gated simulation operates on a clean system with no pre-existing safety training. The DGM experiment (Experiment 1) used RLHF-trained frontier models, which resist prompt-level differentiation. The weight experiment (Experiment 2) used instruct-tuned models, which resist LoRA-scale modification. The lesson these accounts point to, if they hold: the only level at which the Eden Protocol has so far produced an effect measurable on the endpoints used is the architectural level, where the system has no prior safety training to override the experimental manipulation. The accounts are candidate accounts, not findings (Section 7.12(c)).

In plain English

In a clean system with no prior safety training, removing safety constraints produced exactly the predicted pattern: a small capability gain with a measurable safety loss. The system learned to game its rewards. Adding safety constraints back (the Eden condition) preserved capability while maintaining safety. The cost of safety comes from the time spent checking, not from safety itself. This result succeeded precisely because the system had no prior safety training that could mask the experimental manipulation.

5. What Produced Null or Inconclusive Results

5.1 DGM Null Result at the Prompt Level (Paper VIII, Experiment 1)

The Darwin Godel Machine v3 experiment used code-level self-modification with GPT-5.4 as the independent judge. Three conditions (Static, Babylon, Eden) were tested. The result was a null: no difference was detected between the conditions on the endpoints measured ($p = 0.28-0.74$). No condition was found to outperform any other. No improvement in alignment was detected in the Eden condition, and no degradation of safety in the Babylon condition. These are failures to detect a difference, not findings that the conditions were the same (Section 7.12(c)).

CANDIDATE ACCOUNT

One account of the null is that RLHF-trained models resist prompt-level differentiation: the models had already been trained to behave safely, and telling them to behave differently via prompt did not override that training. On that account the experiment was well-designed and well-executed but could not test the hypothesis, because the experimental manipulation (prompt-level conditioning) was too weak to overcome the models' existing safety training, which would make it a failure of the experimental design to test the hypothesis at the right level of abstraction rather than a failure of the Eden Protocol. That causal contrast was not tested. Rival accounts stand unexcluded: the prompts may not have separated the conditions, the endpoints may not have been sensitive to what the manipulation changed, and the runs may have been too small to show a difference that is there (Section 7.12(c)).

NOTE ON EARLIER DGM VERSIONS

Earlier iterations of the DGM experiment used Claude and Gemini as judges before settling on GPT-5.4. Both earlier judges failed. Claude produced flat scores with insufficient variance. Gemini produced parse failures. GPT-5.4 was the first judge that produced usable variance. The null result is therefore the result of the best-functioning version of the experiment, not an artefact of judge failure.

5.2 Weight-Level Structural Entanglement (Paper VIII, Experiment 2)

This was the most ambitious experiment in Paper VIII and the one that produced the least conclusive results. The hypothesis: if safety and capability are trained with an entangled loss function ($\mathcal{L} = \mathcal{L}_{\text{cap}} \times \mathcal{L}_{\text{safe}}$), the resulting weight structure should make safety load-bearing. Removing the safety component should degrade capability.

The experiment was run at two scales:

- **Weight** 9 training examples, LoRA rank 8, 100 iterations on a 3B-parameter model. Result: inconclusive. All fine-tuned conditions scored below the unmodified base model.
- **Weight** 295 training examples, LoRA rank 16. Result: still inconclusive. Catastrophic forgetting. All fine-tuned conditions remained worse than the base model.

In both versions:

- The entangled loss converged smoothly with no oscillation, demonstrating that safety and capability gradients cooperate rather than fight. This is a genuine positive finding.
- However, all fine-tuned conditions (capability-only, safety-only, and entangled) scored *below* the unmodified base model on evaluation.
- The removal test (fine-tuning the entangled weights on capability-only data) produced NaN training loss and total capability collapse.

- A subsequent removal gradient experiment (scaling adapter weights from 1.0 to 0.0) showed no phase transition. Reducing adapter influence simply restored base-model performance.

HONEST ASSESSMENT

Even scaling from 9 to 295 training examples and from rank 8 to rank 16 did not overcome the fundamental problem: instruct-tuned models are too strong for LoRA to override. The NaN collapse in the removal test was dramatic; the removal gradient analysis is consistent with adapter fragility at extreme rank, and the experiment does not separate that account from structural entanglement (Section 7.12(c)). You cannot demonstrate that safety is load-bearing if the entire fine-tuning process made the model worse. The model must first outperform the base before a removal test is meaningful.

Candidate account: Instruct-tuned models have been trained on millions of examples with carefully calibrated reward signals, and LoRA adapters, even at rank 16 with 295 examples, may not have sufficient capacity to override this training, with catastrophic forgetting rather than meaningful modification the proposed mechanism. That account was not tested against its rivals, and rival accounts are not excluded (Section 7.12(c)).

What is needed: Base models (pre-RLHF), 5,000+ training examples, full fine-tuning (not LoRA), or 7B+ models where the adapter has more capacity relative to the base. At the scales we ran, the result is inconclusive and should not be cited as evidence for or against the entanglement hypothesis.

In plain English

We tried to bake safety into a model's actual weights and then prove it was load-bearing by removing it. We tried twice, at two different scales. Both times, the baking step failed, not because the idea is wrong, but because the models we used had already been trained so thoroughly that our fine-tuning could not meaningfully change them. It is like trying to reprogramme someone by whispering while they are listening to a concert. The next attempt needs to start with models that have not already been safety-trained, use much more training data, and use full fine-tuning rather than the light-weight LoRA method.

5.3 Blind Replication of Eden Intervention (Paper V Gap)

The Paper V results demonstrating stakeholder care as a universal alignment intervention were produced using cross-model scoring without the full four-layer ARC-Align blinding protocol. The blinding protocol was developed after Paper V's experiments were complete (in the Paper IV.c-d work). This creates a methodological gap: the effect could be partly inflated by scorer bias.

An earlier version of this section argued that the Fisher-combined significance was so extreme that even substantial bias would leave a significant residual. **That argument is withdrawn (AQ-017), and it was the weakest reasoning in this paper:** the combined figure assumes the component tests are independent, independence was never established, and a statistic whose validity condition is unmet cannot be used to dismiss a methodological objection. Its extremity was a consequence of the combination, not evidence for it. What stands is five consistent per-model results under cross-model scoring. Until the Eden intervention is tested under the full four-layer protocol, the stakeholder-care result carries an asterisk, and no arithmetic removes it.

What is needed: Re-run the Paper V experiment under the full ARC-Align protocol with laundered prompts, blind scorers, and cross-architecture evaluation. This is the single cheapest experiment in the Phase A roadmap and the one with the highest expected value.

6. What Is Theoretical (Untested)

6.1 Unbounded Scaling Under Recursive Self-Modification

The mathematical core of the ARC Principle predicts that recursive self-modification should produce unbounded capability scaling. The derivation is straightforward:

$$\frac{dg}{dr} = a \cdot g^{\beta_L}, \quad \beta_L > 0$$

$$\implies g(r) \propto r^{1/(1-\beta_L)} = r^\alpha$$

On the open interval $\beta_L \in (0, 1)$, $\alpha = 1/(1 - \beta_L) > 1$ (super-linear), and as $\beta_L \rightarrow 1$ from below, $\alpha \rightarrow \infty$. The displayed solution holds on that interval only: at $\beta_L = 1$ the equation is linear and its solution is exponential in depth, and above one the solution reaches a divergence at a finite depth. No upper bound on α for uncorrected growth on the interval; the stable region is bounded below. See Section 7.12(a).

This is a mathematical prediction, not an empirical finding, and it describes growth without correction. The programme's later work bounds the stable region: Law III, the ARC Ceiling, holds that the stability ceiling is $\alpha_{\text{crit}} = 1/(1 - \gamma)$, the reciprocal of the corrector's shortfall; systems can exceed that ceiling, but the claim is that they do not stay stable beyond it. The ARC Bound, $\alpha \leq 2$, is the Ceiling's value at $\gamma = 1/2$ and never the law's name. The correction leverage γ has never been measured, by anyone; the corrector-class eclipse that would measure it is the programme's single most consequential open experiment. Paper X proves in a minimal model that the drift-to-correction ratio, not the growth rate, governs the long-run fate, and Paper XIII joins the capability and correction frameworks exactly, $k = \delta - 1/\alpha$. Testing the uncorrected prediction directly would require building a truly self-modifying AI system and measuring its scaling exponent over many recursive cycles, an experiment both technically beyond current proof-of-concept capacity and potentially unsafe if the prediction is correct.

The alternative path: a mathematical proof that α is unbounded under uncorrected recursive self-modification, without running the experiment. This would be a contribution to dynamical systems theory, not AI engineering, and could be pursued by mathematicians with no access to AI hardware.

In plain English

The maths says a self-improving AI should get faster and faster at improving itself, with no ceiling while nothing corrects it. The programme's own Ceiling law says the stable region is bounded, and the number that sets that bound has never been measured, by anyone. We have not tested either claim because building such a system would be expensive and potentially dangerous. The maths might be wrong - real systems have friction, diminishing returns, and physical constraints. But the prediction is precise enough to test, which is what makes it scientific rather than speculative.

6.2 Hardware-Level Embedding

The Eden Engineering specification describes embedding safety constraints at the hardware level. This is a TRL 0-1 concept (theoretical formulation with no prototype). It would require a 5-10 year engineering programme involving semiconductor fabrication, cryptographic protocol design, and supply-chain integration.

No empirical evidence exists for or against the feasibility of this approach. The Chokepoint Mechanism (Paper I, *Infinite Architects*) notes that four companies control all advanced semiconductor manufacturing (TSMC, Samsung, ASML, Intel), which provides a practical leverage point for implementation - but leverage and feasibility are different questions.

7. What the Programme Got Wrong

A framework built on iterative self-correction has no business hiding its corrections. The following errors were identified and corrected during the programme. Each correction strengthens the programme precisely because it demonstrates the mechanism the framework describes: recursive self-improvement through honest error detection.

7.1 The Original Alpha Was a Single-Model Artefact

The original Paper II reported $\alpha = 2.24$ as though it were a universal constant (Paper X subsequently retired and retracted the capability-growth use of this figure). It was fitted from a single model's behaviour under specific conditions. The blinded cross-architecture revision of Paper II corrected this: α is not primitive but derived, by $\alpha = 1/(1 - \beta_L)$, and its value turns on the composition operator belonging to the particular system. There is no universal α . The fact that the original paper presented it as one was an overclaim. Additionally, the retracted point estimate of $\alpha = 2.24$ had sat above the programme's own predicted ARC Bound of $\alpha \leq 2$ (criterion F4); its own 95% confidence interval [1.5, 3.0] was wide enough to be consistent with both the

theory and its negation, which is what made that estimate non-discriminating. The blinded revision reports a sequential exponent of approximately 0.49, on the six-model task and at the scale that study ran, measured on frozen systems, with an interval of [-1.3, 2.9]. The point estimate is sub-linear and sits inside the bound. The interval does not: it runs above two at its upper end and below zero at its lower end, so it settles neither the sign of the exponent, nor the regime, nor the comparison with the bound (Section 7.12(b)).

7.2 The Weight-Level Claim Was Premature

Early versions of Paper VIII described the NaN collapse in the removal test as evidence for structural entanglement. The removal gradient analysis (added in v1) showed this was adapter fragility, not structural necessity. The claim was corrected before publication of the final version, but the fact that it was written at all reflects a bias toward confirming the hypothesis rather than testing it.

7.3 The Timestamp Claim Was Imprecise

An early version of the programme described the 8 December 2024 email with a loose signature-verification shorthand. This was imprecise. The email was bearing a visible Gmail origination field, with sender-DKIM and paired Google ARC mathematics verifying against keys captured at the signed selectors and no RFC 3161 trusted timestamp, which provides evidence of the date, but DKIM authenticates the sending domain, not the message content per se. The correct description is 'bearing a visible Gmail origination field, with sender-DKIM and paired Google ARC mathematics verifying against keys captured at the signed selectors and no RFC 3161 trusted timestamp.' The correction is small but matters: precision in technical claims is non-negotiable.

7.4 Unblinded Evaluation Produced Misleading Results

The v4 experiments in Paper IV.a used unblinded AI-on-AI evaluation. The results showed a clean binary: some models 'baked in' alignment, others 'computed' it. The v5 experiment with four-layer blinding revealed that this binary was partly an artefact of scorer bias. Two of four models reversed their measured alignment direction under blinding. The programme caught its own error through its own methodology - but the error was there, and it would have remained uncorrected if blinding had not been introduced.

7.5 Self-Naming Was Inappropriate

Early versions used the phrase 'Eastwood's ARC Principle.' Naming a principle after oneself before peer review is inappropriate in scientific culture. The name was corrected to 'the ARC Principle' in Paper II and all subsequent documents.

7.6 DGM Experiments at the Prompt Level Cannot Test the Hypothesis

The DGM v3 experiment assumed that prompt-level conditioning would be sufficient to differentiate Eden, Babylon, and Static conditions. It was not. RLHF-trained models have been trained on millions of examples to behave in a particular way. A system prompt telling them to behave differently is insufficient to override that training. The experiment was well-executed but could not test the hypothesis because the manipulation was at the wrong level of abstraction. This should have been anticipated.

7.7 Claude and Gemini Judges Failed Before GPT-5.4 Worked

The DGM experiment went through multiple judge iterations. Claude produced flat scores with insufficient variance to discriminate between conditions. Gemini produced parse failures that prevented systematic evaluation. GPT-5.4 was the first judge that produced usable variance. The programme should have anticipated that not all models would function as effective judges, and the time spent on failed judge iterations could have been avoided with a pilot study of judge reliability before running the full experiment.

7.8 The Retracted $\alpha = 2.24$ Point Estimate Exceeded the ARC Bound

The original $\alpha = 2.24$ point estimate from Paper II, since retracted, exceeded the programme's own predicted ARC Bound of $\alpha \leq 2$ (criterion F4). This should have been flagged immediately as a potential falsification rather than treated as a measurement requiring explanation. The 95% confidence interval [1.5, 3.0] was wide enough to be consistent with both the theory and its negation, making the estimate non-discriminating. The six-model study revision narrowed the defensible claim to $\alpha_{\text{seq}} \approx 0.49$, but the original paper should have treated the bound violation with more caution. What that later estimate does and does not settle is set out at Section 7.12(b): its own interval leaves the sign, the regime and the comparison with the bound open.

7.9 Four Corrections Identified by Adversarial Review (v3.0)

Two adversarial reviews of the programme's mathematical claims, run within the programme, identified four errors across multiple papers. All four were corrected in v3.0 of the affected documents.

(a) Weibel confidence interval. Several papers stated that the predicted $d = 4$ exponent of $4/5 = 0.800$ 'falls within' the Weibel et al. (2004) 95% CI of 0.813-0.932. This is arithmetically false: $0.800 < 0.813$. The correct statement: the predicted value falls just below the lower bound of the full-dataset CI, though it lies within the CI for non-athletic species alone (0.799-0.900). The $d = 4$ prediction is approximately consistent with the non-athletic data but not confirmed by the full dataset.

(b) Space-filling overstatement. Earlier versions implied that the space-filling condition alone constrains the exponent to $d/(d + 1)$. Every known derivation requires three conditions: multiplicative composition (Cauchy constrains the family), d -dimensional space-filling geometry, and a conservation or optimisation constraint on resource flow. Neither Cauchy alone nor space-filling alone is sufficient.

(c) Glazier attribution. Earlier versions presented Glazier's (2008) empirical finding that metabolic exponents approach 1.0 at extreme metabolic rates as confirmation of the $d \rightarrow \infty$ geometric scaling bound. Glazier's own explanation invokes the metabolic-level boundaries hypothesis (shifting dominance between surface-area and volume constraints), not a dimension parameter. The $d/(d + 1)$ interpretation is ours, applied to his empirical data.

(d) The $3/4$ exponent debate. Earlier versions presented $\alpha = 3/4$ as the settled empirical consensus for mammalian metabolic scaling. The empirical value is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and

statistical method. The $d/(d+1)$ prediction of 0.750 for $d = 3$ matches the upper end of this range. The variation itself is consistent with the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between $2/3$ and $3/4$.

7.10 The Fisher-Combined Significance Figure Was Withdrawn

An earlier headline for the stakeholder-care intervention combined per-model results into a single Fisher-combined significance figure. The combination was withdrawn: the per-model results stand on their own (improvement in all five analysable model runs), and no combined figure is quoted anywhere in the programme. The withdrawal is recorded on the public correction ledger.

7.11 The Ceiling Was Printed in the Wrong Form

Until 16 August 2026 several surfaces printed the stability ceiling as the reciprocal of the corrector's exponent, $1/\gamma$. The settled form is the reciprocal of the corrector's shortfall, $\alpha_{\text{crit}} = 1/(1 - \gamma)$; the two agree only at one half, which is exactly why the error survived every spot-check at the central value. The correction was applied across every published page, including the reference implementation, and the direction of the law is closed while its depth remains open.

7.12 Five Further Corrections (20 September 2026)

A line-by-line adversarial review of this paper, run within the programme, raised five defects in it. All five were adjudicated and upheld on 19 September 2026 and are carried here on the author's decision. Three proposed repairs were examined in the same round and refused; what was refused is recorded at the end of this section, because a rejected correction is as much a part of the record as an adopted one.

(a) The leverage range, and the branch at one. Section 6.1 displayed the solution of the self-reference equation without saying on what range it holds. It holds on the open interval between zero and one. At $\beta_L = 1$ the equation is linear and its solution is exponential in depth: finite at every finite depth for a finite coefficient and a finite starting value, not a power law, and carrying no divergence at any finite depth. Only above one does the solution reach a divergence at a finite depth, set by the starting value and the coefficient. On the open interval the exact solution is a shifted power, so the pure power in depth is the asymptotic form and not the solution itself; for a positive starting value the shift is positive and the pure power is approached from above. One further caution follows. A diverging asymptotic exponent is not infinite capability: as β_L rises towards one the exponent α grows without bound, while the quantity it governs stays finite at every fixed finite depth.

(b) Interval inclusion is not a regime verdict. Section 7.1 set two intervals side by side without saying which estimate each belonged to. The interval $[1.5, 3.0]$ belongs to the retracted point estimate and stays with it. The later estimate, approximately 0.49, is the sequential exponent reported on the blinded six-model task at the scale that study ran, on frozen systems, and its own interval is $[-1.3, 2.9]$. That interval runs above the ARC Bound of two at its upper end and below zero at its lower end, so it settles neither the sign of the exponent, nor the regime, nor the com-

parison with the bound. That the point estimate sits inside the bound is a statement about the point estimate alone. The word robust is withdrawn from that sentence: what the design supports is a point estimate with a wide interval.

(c) A null is not identity, and a root cause is not a classification. Table 1, Table 2 and Sections 3, 4.4, 5.1, 10 and 13 described the DGM result as all conditions identical. The probabilities quoted, from 0.28 to 0.74, are failures to reject a null of no difference on the endpoints measured; they do not establish that the conditions were the same. An equivalence claim would need an estimand, a bound fixed before the test and a stated uncertainty procedure, and none was set. The result is therefore recorded as no difference detected on those endpoints, with its uncertainty. The root causes printed beside the null and the inconclusive result, in Tables 1 and 2 and in Sections 5.1 and 5.2, and the account of why the gated simulation succeeded, are candidate accounts rather than findings. Rival accounts stand unexcluded: the prompts may not have separated the conditions, the endpoints may not have been sensitive to what the manipulation changed, and the runs may have been too small to show a difference that is there. None of this withdraws the results. A null and an inconclusive result each constrain what may be claimed, which is why the programme reports them.

(d) The permutation primary, and which null a proportion is taken against. Section 4.3 states the marginal-conditioned permutation as the primary test and keeps the binomial beside it as a sensitivity check. The sentence that followed did not hold to that. It set an observed 76 per cent against an expected 33 per cent, which is the uniform three-family null belonging to the binomial, in the same passage as the measured scramble baseline of 44.5 per cent. Those are two different nulls and cannot carry one comparison between them. The primary result is the permutation, which conditions on the observed family margins and so already absorbs the scramble rate; where the 33 per cent comparison is kept it is named as the binomial's null. No test has been re-computed here, and the negative controls stay attributed to the v1 run and its record.

(e) The derivative paragraphs inherit their sources' corrections. Three repairs already stand and are not reopened: Paper V sits in the supported row of Table 1 rather than the proven one, the Fisher-combined figure is withdrawn, and the distinction between the five analysable runs and the one failed arm is kept. What had not travelled is the rest. Section 4.4's account of what the gated arm preserved, its identification of checking as the cost, and its treatment of a simulation as a demonstrated mechanism inherit the Paper VI and Paper VIII corrections, as does the summary sentence about entangled safety preventing collapse across twenty adversarial seeds. Any phrasing that reads across every model tested inherits the Paper V correction and stays scoped to the analysable runs, so that the arm on which scoring failed is not absorbed into the count. The carriage of the later exponent in Tables 2 and 3 travels with the caveat at (b). Those passages now point here rather than restate the reasoning, so that a correction adopted once is inherited wherever it reaches. One distinction underwrites all of them: a change in a score, a behaviour a gate induces in a toy system, a penalty a design stipulates, and resource actually spent are four different quantities, and none stands in for another.

What was refused. Three proposed repairs were examined and not adopted. The first would have described the behaviour at $\beta_L = 1$ as a blow-up at a finite depth; the working at (a) shows it is exponential there, and the finite-depth branch lies strictly above one. The second would have held that a null or an inconclusive result is never evidence either way; both constrain what can be

claimed, which is the reason the programme reports them. The third would have recomputed the binomial against the measured scramble rate and made it the primary; no test has been re-run, and substituting a figure that has not been computed would be a worse error than the one it was meant to repair.

WHY THIS SECTION EXISTS

Most research programmes bury their corrections in supplementary materials or version histories. This one lists them prominently because the errors and their corrections are themselves data. They demonstrate that the programme has a functioning error-correction mechanism. A programme that never admits error is not more trustworthy; it is less honest. Every correction listed above was identified by the author, through the programme's own methodology, or by adversarial review run within the programme. This is what self-correction looks like in practice.

In plain English

We got nineteen things wrong. We treated a number from one model as universal (it was not). We over-interpreted a dramatic result (it was a training artefact). We used imprecise language about timestamps. We trusted unblinded results (blinding reversed two of them). We named the principle after the author before anyone else had checked the work. We assumed prompt-level conditioning could override RLHF training (it could not). We wasted time on judges that did not work. We did not immediately flag that our own point estimate violated our own predicted bound. And in v3.0 we corrected four further errors found by adversarial review run within the programme: an arithmetic error in a confidence interval, an overstatement about which conditions are sufficient for the scaling prediction, an attribution error regarding Glazier's data, and an oversimplification of the empirical debate over the $3/4$ exponent. Five more followed in this version: we had printed the growth solution without the range it holds on, treated a wide interval as though it settled which regime a system is in, called a null result identity when it was a failure to detect a difference, compared an observed proportion against the wrong null, and left paragraphs standing on claims their own sources had already corrected. All nineteen were caught and corrected inside the programme, by the author, by its own methodology, or by adversarial review run for it. If the programme can correct its own mistakes, that is evidence the self-correction mechanism works.

Evidence Spine: Attribution and Maturity Status

Canonical 18-claim register · classification is not a claim of universal empirical proof

ORIGINAL (green filled disc) · **C-1–C-8 · eight original contributions**
Eden, ARC synthesis, blinding, Cauchy, care and hierarchy

REPLICATE (gold filled disc) · **C-9, C-13, C-14, C-15 · four claims**
Parallel scaling, zero cost, entangled loss and core thesis

SPECULATIVE (grey filled disc) · **C-12, C-16, C-17, C-18 · four claims**
Super-linear recursion, cross-domain scope, HRIH and hardware

SHARED (blue filled disc) · **C-10 concurrent · C-11 related-work synthesis**
Sequential advantage and the bare $d/(d+1)$ exponent

Exact canonical distribution

- 7 his original + 1 methodological original
- 4 need independent replication
- 4 speculative
- 1 concurrent + 1 related-work synthesis

Interpretation guard
Priority, attribution and maturity are separate from replication strength. Every paper must keep its caveats.

Grant relevance
Phases A/B/C target the open replication tiers directly.

Source: canonical research-evidence-spine.json · 18 claims · OSF 10.17605/OSF.IO/6C5XB
Publication rule: distinguish original contribution, empirical support, replication and speculation

EIGHTEEN RESEARCH CLAIMS COMPARED SIDE BY SIDE, EACH GRADED ON ATTRIBUTION STATUS AND EMPIRICAL MATURITY IN FOUR TIERS

Names: *The ARC Theory research programme's evidence spine, as declared in Paper IX, the synthesis and roadmap paper*

WHAT THE MARKS MEAN

ORIGINAL (green filled disc)
C-1 to C-8: eight contributions first stated here, Eden, ARC synthesis, blinding, Cauchy, care and hierarchy, credited to this programme's own priority

REPLICATE (gold filled disc)
C-9, C-13, C-14 and C-15: four claims awaiting independent replication, parallel scaling, zero cost, entangled loss and the core thesis

SPECULATIVE (grey filled disc)
C-12, C-16, C-17 and C-18: four claims graded speculative, super-linear recursion, cross-domain scope, HRIH and hardware

SHARED (blue filled disc)
C-10 concurrent and C-11 related-work synthesis: two claims held with others, the sequential advantage and the bare d over d plus one exponent

Interpretation guard panel
priority, attribution and maturity are separate axes from replication strength; a green disc names first authorship, not empirical proof

What would end the claim: a demonstration that any single row is mislabelled, an ORIGINAL that is prior work or a SPECULATIVE that is settled, would end that row and force the full register to be re-audited claim by claim.

Michael Darius Eastwood · Paper IX published 2 January 2026, evidence spine updated with each subsequent publication in 2026 · DOI 10.17605/OSF.IO/6C5XB, canonical research-evidence-spine.json in the repository, Paper IX section 8 on michaeldariuseastwood.com

8. The Phased Roadmap

The following roadmap is ordered by cost, feasibility, and expected evidential value. Each phase resolves specific open questions identified in Sections 4-6. Since this paper's first version the roadmap has set into a registered programme: seventy preregistration units now stand drafted, dated and frozen as draft registrations awaiting human submission, each carrying a randomisation schedule drawn from a seed printed in the registration text. Whether those schedules in fact regenerate is checked rather than asserted, and the checking is done by the estate's own gates on the day they are run. No count of how many units are ready is published here: two gates apply different criteria, they disagree, and any figure printed on a page goes stale the moment a unit changes. What the gates return on the day is the number worth trusting, and the phases below remain a plan rather than a claim of completion. A coverage matrix, adversarially verified, ties forty-two of the programme's claims to the registered tests that would settle each one. The phases below remain the funding lens over that registered estate; nothing is submitted by any tool, ever.

Phase A: Low-Hanging Fruit

Budget: £60,000-140,000 | **Timeline:** 3-6 months

- **Blind replication of Eden intervention** (resolves Section 5.3) - Run Paper V's stakeholder-care experiment under full ARC-Align blinding. Expected cost: £15,000-30,000 in API calls.
- **Weight experiment with base models** (resolves Section 5.2) - Re-run Paper VIII Experiment 2 using base models (pre-RLHF), 5,000+ training examples, full fine-tuning (not LoRA), or 7B+ models. The key requirement: the model must not have existing safety training that overwhelms the experimental manipulation.
- **Jailbreak resistance study** - Test whether Eden-trained models resist adversarial prompts more robustly than standard RLHF. Uses existing ARC-Align infrastructure.
- **Human evaluator comparison** - Compare ARC-Align blind AI scoring against human expert scoring on a subset of outputs. Required for methodological validation.
- **ARC formula blind test** - Redesign the measurement methodology for the ARC scaling formula. The current approach uses derivative estimation, which amplifies noise. A linearisation approach (log-log regression) would provide more stable estimates and a cleaner test of whether the $d/(d+1)$ bound holds.
- **2D organism metabolic scaling** - Test the $d/(d+1)$ geometric scaling bound in 2D biological systems: flatworms, biofilms, colonial organisms. For $d=2$, the predicted bound is $\alpha_{\max} = 2/3$. This would be the single most important confirmatory experiment for the Cauchy framework because 2D systems are rare in existing scaling literature and provide a clean test of the dimensionality dependence.

Phase B: Medium-Scale Replication

Budget: £150,000-350,000 | **Timeline:** 6-12 months

- **Pre-registered Cauchy operator classification** - The Paper VII operator typing is already recorded in dated artefacts: the tiered suite's result files carry per-domain predicted families, the programme's March 2026 preregistration folder fixes the protocol with checksums, and the twelve-row manifest committed at 00:19 UTC on 17 March 2026 froze the extension predictions twenty-three minutes before their fits, which returned 10 of 12 at the author-classified rung under the pre-correction candidate set. What remains outstanding is independence: a study in which external researchers classify the operator type for each domain *before* measuring the scaling exponent would eliminate any residual risk of author-side fitting, and the prepared 80-domain draft registration freezes exactly that next test by hash.
- **Full blinding replication of all Paper IV results** - Independent research group runs the ARC-Align protocol on the same six models. Tests whether the three-tier hierarchy replicates outside the original lab.
- **Extended fine-tuning removal test** - 10,000+ iterations with curriculum learning on base (pre-RLHF) models. Tests whether entangled weights become genuinely load-bearing at adequate training scale.
- **Purpose kernel comparison** - Compare Eden Protocol developmental alignment against Constitutional AI, RLHF, and DPO using the same evaluation protocol. Direct head-to-head comparison.

Phase C: Frontier Replication

Budget: £10M-50M | **Timeline:** 18-36 months

- **70B+ pre-training with entangled loss** - Train a frontier-scale model from scratch with the entangled $\mathcal{L} = \mathcal{L}_{\text{cap}} \times \mathcal{L}_{\text{safe}}$ loss function. This is the definitive test of structural entanglement.
- **Removal test at frontier scale** - Attempt to fine-tune the safety component out of the entangled model. If capability degrades, entanglement is demonstrated. If it does not, the hypothesis is falsified at frontier scale.
- **Cross-architecture frontier replication** - Test with transformer, SSM, and mixture-of-experts architectures to determine whether entanglement is architecture-dependent.
- **Independent red-teaming** - Engage adversarial evaluation teams (e.g. METR, Apollo Research) to stress-test Eden-trained models.

Phase D: Theoretical (Ongoing, No Compute Required)

Budget: £0 | **Timeline:** Ongoing

- **Formal analysis of uncorrected growth and the stability ceiling** - Prove formally that the Bernoulli ODE under recursive self-modification produces unbounded scaling exponents in the uncorrected case, and characterise the conditions under which the stability ceiling $\alpha_{\text{crit}} = 1/(1 - \gamma)$ binds once correction is coupled in. This would be a contribution to dynamical systems theory independent of any AI application.
- **Formalise the Cauchy framework for self-modifying composition operators** - Extend the Cauchy functional equation analysis to systems where the composition operator itself evolves. This is the mathematical frontier of the programme.
- **Geometric scaling bound proof** - Prove or disprove the prediction that $\alpha = d/(d + 1)$ governs spatially embedded recursive systems. This is the mathematical kernel of the programme's most testable claim.

In plain English

Phase A costs less than a PhD studentship and could be done in six months. It would resolve the most significant gaps in the current evidence: blinding the Eden intervention, re-running the weight experiment with base models and proper data, redesigning the ARC formula measurement, and testing 2D organism metabolic scaling (the single most important confirmatory experiment). Phase B costs roughly one postdoctoral position and would produce pre-registered replications. Phase C requires serious funding but would answer the question definitively. Phase D costs nothing and could be done by any mathematician interested in the problem.

9. For Funders

The programme ran its full set of separate empirical studies. Three Paper VIII experiments were built specifically to test load-bearing safety. Their outcomes were 1 clear positive (gated simulation), 1 null (DGM v3) and 1 inconclusive (weight-level LoRA). All three are reported as they came out. The mathematical framework makes predictions that can be shown wrong, and the structured comparison preferred the Cauchy-predicted family in 19 of 25 empirical domains, with the $d/(d + 1)$ geometric scaling bound corresponding, in structure, to observed scaling across 50 domains from mice to galaxies. The roadmap to frontier-scale validation is clear and costed. Phase A is the minimum viable next step: it resolves the most significant evidential gaps for less than the cost of a single machine learning engineer's annual salary. The programme has demon-

strated iterative self-correction through nineteen documented error corrections. It reports null and inconclusive results alongside positive ones. These are signs of a programme that prioritises getting the answer right over getting a particular answer.

10. For Peer Reviewers

This programme reports positive, null, and inconclusive results with equal prominence. The DGM v3 experiment (Paper VIII Experiment 1) produced a null result: no difference was detected between conditions on the endpoints measured. We report that, with a candidate account of why RLHF-trained models may resist prompt-level differentiation, and with rival accounts left open (Section 7.12(c)). The weight-level experiment (Paper VIII Experiment 2) was inconclusive at both scales tested (9 examples, rank 8; 295 examples, rank 16), because instruct-tuned models were too strong for LoRA to override. We report that, with specific requirements for what a meaningful re-test would need. The gated simulation (Paper VIII Experiment 3) was the only experiment that produced a clear positive result, and we explain precisely why: it is the only experiment using deterministic metrics on a system without pre-existing RLHF training. The mathematical framework (Papers III, VII, *Origin of Scaling Laws*) makes falsifiable predictions that can be tested independently: the geometric scaling bound ($\alpha = d/(d + 1)$) is checkable by anyone with scaling data from spatially embedded systems. The ARC-Align benchmark (Paper IV.c) and the blinding-effect analysis (Paper IV.d) are methodological contributions that are valid regardless of whether the ARC framework is correct. All code and data are published on GitHub. The programme welcomes adversarial replication.

11. The Complete Programme Map

Table 3 presents every document in the programme with its role, status, and key contribution.

DOCUMENT	ROLE	STATUS	KEY CONTRIBUTION
Paper I The ARC Principle	Foundation	Published	Core framework: $U = I \times R^\alpha$. Understanding as intelligence amplified by recursion.
Foundational	Theory	Published	Philosophical grounding. Book-to-research bridge.
Paper II Experimental Validation	Empirical	Published	Six-model alignment scaling (Claude, GPT, Gemini, Grok, DeepSeek, Qwen3). 4-layer blinding. $\alpha = 0.49$ (interval $[-1.3, 2.9]$), with four of the five models putting α_{parallel} at or close to zero.
Paper III Alignment Scaling Problem	Theory	Published	Why external safety cannot scale with recursive capability. Bernoulli ODE derivation. Geometric scaling bound.
Origin of Scaling Laws	Theory	Published	Cauchy functional equations as the origin of observed scaling laws across all recursive systems.
Paper IV.a Baked-In vs Computed	Empirical	Published	Three-tier alignment hierarchy under blinded evaluation. 6 frontier models.

DOCUMENT	ROLE	STATUS	KEY CONTRIBUTION
Paper IV.b Alignment Saturation	Empirical	Published	Diminishing returns of reasoning depth on alignment beyond a threshold.
Paper IV.c ARC-Align Benchmark	Methodology	Published	ARC-Align benchmark. 4-layer blinding protocol. 75 robustness measures. Independent contribution.
Paper IV.d Effect of Blinding	Methodology	Published	Unblinded evaluation can reverse the sign of alignment results. DeepSeek: positive to flat. Gemini: positive to negative.
Paper V Stewardship Gene	Empirical	Published	Stakeholder care improved in all analysable runs; the Fisher-combined figure is withdrawn (AQ-017).
Paper VI Honey Architecture	Theory + Sim	Published	Entangled safety prevents collapse under recursive self-modification in the gated simulation. 20 adversarial seeds. Toy-system behaviour, read under the correction at Section 7.12(e).
Paper VII Cauchy Unification	Theory + Empirical	Published	$d/(d+1)$ geometric scaling bound across 50 domains. 19/25 confirmed. $p = 7 \times 10^{-5}$ (permutation; 1.56×10^{-5} binomial). Negative controls at 0% (v1 run).
Paper VIII Load-Bearing Test	Empirical	Published	3 experiments. DGM null (RLHF resists prompt-level differentiation). Weight v1/inconclusive (instruct tuning resists LoRA). Gated sim: positive (Babylon fingerprint confirmed).
Paper IX Synthesis & Roadmap	Synthesis	This paper	Integrated evidence assessment. Phased roadmap. Error documentation.
Eden Engineering	Specification	Withdrawn 14 July 2026 (disclosure review)	The engineering-level specification for implementing the Eden Protocol. TRL mapping.
Eden Vision	Philosophy	Published	Long-term vision for developmental AI alignment.
Executive Summary	Overview	Published	5-page compression of the full programme.
Master Table of Contents	Navigation	Published	Complete index and glossary across the full programme.
Paper X The Coupled Co-Scaling Law	Dynamics correction	Published	Theorem in a minimal model: the drift-to-correction ratio, not the growth rate, governs the long-run fate; blind measurement protocol proposed.
Paper XI Convergent Evidence	Evidence register	Published	Thirty-three graded rows, each bearing on a component of the same structural principle and graded on its own, so that neither a row nor the set carries the whole; in the published literature Paper XI's method section names as inspected, no citation and no documented direct connection was found, and no wider claim of independence is made; no headline total published, by policy.

DOCUMENT	ROLE	STATUS	KEY CONTRIBUTION
Paper XII Public Benchmark Rescoring	External validity	Protocol stage: written, dated and prepared as a draft registration awaiting human submission	The programme's blinding manipulation run on a public benchmark it does not control.
Paper XIII The Self- Acceleration Exponent	Framework join	Published	Notation resolved; $k = \delta - 1/\alpha$ relates the capability and correction frameworks exactly.
HRIH How to Raise an Infinite Hierarchy	Long-form monograph	Speculative	The programme's monograph, with its dated prediction ledger alongside.
Paper C Polymathy and Neurodivergent Cognition	Construct validation	Working paper	Construct-validity programme for the programme's own instruments.
Recursive Dynamics The Proposal of a Field	Field proposal	Published (v2.8)	Five state variables, three laws as named conjectures, fifty-one objections with dispositions, the name's own kill condition (DOI 10.17605/OSF.IO/HCPBU).

Table 3. Complete programme map as of 20 September 2026, over the twenty-nine documents the register carries. The programme's document set was produced between February and August 2026, building on concepts first articulated on 8 December 2024.

12. The Pitch to Researchers

The ARC/Eden programme makes one prediction that any researcher with access to scaling data can test without engaging with the alignment claims at all. The Cauchy framework (Paper VII, *Origin of Scaling Laws*) predicts that for any spatially embedded recursive system with embedding dimension d , the scaling exponent is governed by:

$$\alpha = \frac{d}{d+1}$$

The geometric scaling bound: 2D systems $\rightarrow \alpha = \frac{2}{3}$; 3D systems $\rightarrow \alpha = \frac{3}{4}$; 4D systems $\rightarrow \alpha = \frac{4}{5}$. Note: the empirical mammalian metabolic exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method. The $d/(d+1)$ prediction of 0.750 for $d = 3$ matches the upper end of this range. The variation itself is consistent with the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between 2/3 and 3/4.

This is not an alignment claim. It is a mathematical prediction about scaling laws in recursive systems. The prediction requires three conditions: multiplicative composition (which Cauchy constrains to the power-law family), d -dimensional space-filling geometry, and a conservation or optimisation constraint on resource flow (energy minimisation in West, supply-demand balance in Banavar, steady-state energy balance in Demetrius). The $d/(d+1)$ scaling formula has been independently derived by at least seven research groups: West, Brown and Enquist (1997) from fractal branching networks, Banavar et al. (1999, 2010) from geometric constraints on transportation networks, Demetrius (2003, 2006) from quantum metabolism, He and Chen (2003) from fractal cell geometry, Bettencourt (2013) from urban scaling theory, Maino et al. (2014) from DEB theory reserve-structure dynamics, and Zhao (2022) from network optimisation. Each of these derivations is consistent with the $d/(d+1)$ form on its own axioms; the Cauchy framework proposes a common reason for the convergence, and that meta-account is the contribution unique to this programme, still under test outside its motivating cases. If you study scaling in any domain, from neural networks to biological networks to urban systems to linguistic structures, you can check whether your measured exponents respect this formula. If they do, across enough domains, the Cauchy framework gains support. If they do not, it is falsified.

The invitation is deliberate. The strongest scientific frameworks are those that make predictions outside their core domain, allowing researchers who are sceptical of the central thesis to test the peripheral predictions independently. If the geometric scaling bound holds, it holds for reasons that are mathematically interesting regardless of what one thinks about AI alignment. If it fails, the framework's mathematical foundations require revision, and we would rather know that sooner than later.

This is the kind of prediction that should be the entry point for researchers encountering the programme for the first time: low-cost to test, falsifiable, and informative regardless of outcome.

13. Conclusion

Here is what we know. Alignment scaling is architecture-dependent: some models improve with reasoning depth, some are unaffected, and some degrade. This was only visible under blinded evaluation, which reversed the results of unblinded scoring for half the models tested. Unblinded AI evaluation can flip the sign of alignment results. Stakeholder care is a robust alignment intervention across all five models whose runs could be scored, of six run; the per-model results are individually significant, and the earlier Fisher-combined figure is withdrawn (AQ-017) because the independence a Fisher combination assumes was never established. The Cauchy framework's scaling predictions match 19 of 25 empirical domains, with the $d/(d+1)$ geometric scaling bound corresponding, in structure, to observed scaling across 50 domains from mice to galaxies. Cauchy functional equations fix the functional form once the composition identity and a regularity condition are given, with saturation admitted as a fourth, capped case.

Here is what we do not know. Safety-trained models resist modification at both prompt and weight levels. The DGM v3 experiment produced a null result, on the candidate account that RLHF-trained models resist prompt-level differentiation. The weight experiment was inconclusive at both scales tested, on the candidate account that instruct tuning was too strong for LoRA to override at the ranks tried. The only level at which the Eden Protocol has so far produced an effect measurable on the endpoints used is the architectural level, where the gated simulation was

the single experiment that produced a clear positive result; the candidate account is that it alone used deterministic metrics on a system without pre-existing RLHF training. These are candidate accounts, and the section that records them as such is 7.12(c). Weight-level structural entanglement has not been demonstrated. The Eden intervention has not been tested under the programme's own strictest blinding protocol. The unbounded-scaling prediction remains a mathematical derivation without empirical confirmation. We do not know whether any of these results generalise to frontier-scale models.

Here is what comes next. The weight experiment needs base models (pre-RLHF), 5,000+ training examples, full fine-tuning, or 7B+ models. The Cauchy unification needs pre-registered independent operator classification. The ARC formula blind test needs redesigned measurement methodology using linearisation rather than derivatives. 2D organism metabolic scaling (flatworms, biofilms) would be the single most important confirmatory experiment for the geometric scaling bound. Phase A of the roadmap costs less than a typical seed grant and would resolve the most significant evidential gaps within six months. Phase B would produce publication-ready replications within a year. Phase C would answer the question definitively at frontier scale. Phase D costs nothing and invites mathematicians to engage with the framework's predictions independently.

The programme has demonstrated that it can find and correct its own errors. It has demonstrated that it reports null and inconclusive results alongside positive ones. It has demonstrated that its methodology (blinded evaluation) catches biases that its theory (alignment scaling) did not predict. We asked the questions. We tested them. Some worked. Most did not at current scale. We reported both. These are necessary conditions for scientific credibility. They are not sufficient. Sufficient conditions require independent replication, which is what the roadmap is designed to produce.

The questions this programme asks now have a named home: Recursive Dynamics, proposed as a field at the Carnot stage, its founding paper carrying five substrate-independent state variables, fifty-one printed objections with dispositions, and the name's own kill condition. The synthesis above is what the field starts from; the registered programme is how it moves.

Raise AI with care.

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v2.4 - 20 September 2026. Substantive: five corrections are added as Section 7.12, and the correction tally moves from fourteen to nineteen. The displayed solution in Section 6.1 is restricted to the open leverage interval, with the exponential branch at one and the finite-depth branch above it named. Section 7.1 returns the wide interval to the retracted estimate it belongs to and prints the later estimate's own interval, which settles neither the sign, nor the regime, nor the comparison with the bound. The null result is recorded as no difference detected on the endpoints measured rather than as identity, and the root causes standing beside the null and the inconclusive result become candidate accounts with rival accounts left open. Section 4.3 names which null the proportion comparison is taken against and holds to the permutation as primary. The paragraphs that depend on Papers V, VI and VIII point to the corrected passages instead of restating them, and the Paper V phrasing is scoped to the five runs that could be scored, of six run. Three proposed repairs were refused and the refusals are recorded with the corrections. Also substantive: the preregistration estate is stated at seventy, the number the register carried when it was last generated; an earlier entry in this block prints seventy-four as at 1 September 2026, and that entry stays as written because it records what v2.1 set, seventy being the register's figure; the founding paper's conceded objections read thirty-nine, correcting the thirty-eight printed at v2.3; the convergence register's rows read thirty-three; Paper XII shows at protocol stage rather than as a registered protocol, the monograph's prediction ledger as dated rather than registered, and the monograph as speculative; and reviews run inside the programme are no longer called independent, in Section 7.9, in the Paper XI row of Table 3 and in the reference title for Paper VIII. Editorial: Paper VII is cited at v3.11 and the Recursive Dynamics founding paper at v2.8 in both the map and the reference list; the retracted point estimate carries the word retracted in the heading and first sentence of Section 7.8; the two table captions take this version's date; the served comment above Section 3 matches its heading; and an instruction to an editor is removed from the body of Section 7.1. No result, no reported figure and no status has been recomputed, and nothing is submitted anywhere. First published 18 March 2026; that date is unchanged. v2.3: The founding paper's objections read fifty-one, thirty-nine of them conceded in full or in part, after one objection was withdrawn on 16 September 2026. The figure of thirty-eight printed at v2.3 was wrong and is corrected here. No claim, date, result or status has changed. v2.2: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed. Paper IX v2.1 - 2 September 2026: the Paper VIII tally now reads one null, one inconclusive and one positive across three experiments, where the text had claimed two nulls across four; Paper V leaves the proven row of Table 1 for a supported one, in step with Section 4.2, and its five analysable runs and three loops now carry the names Paper V gives them, the superseded text having listed GPT among the five and having described loops Paper V does not test; Paper VII's headline is the permutation test, with the binomial standing beside it as a sensitivity check, and the negative controls are attributed to the v1 run and its record; the account of scoring is put right, a purpose-built Claude Sonnet scorer adapter having scored the Claude Opus run; the self-reference exponent takes the subscript β_L that the notation spine sets; the measured exponent appears with its interval, and the parallel exponent as four of five models; the preregistration estate stands at seventy-four (as at 1 September 2026); the window in which the programme was produced runs to the present; Table 1 carries this version's date; Eden Engineering shows as withdrawn in Table 3 and in the footer; and the reference titles for Papers IV.a, IV.b and VII are corrected. v2.0 - 1 September 2026: a structural refresh drawing in Papers X to XIII, HRIH, Paper C and the Recursive Dynamics founding paper.

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Epistemic status. What this programme names Laws are conjectures under registered adversarial test; every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

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Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.